

Sales Data Analysis Report

Project Title

Sales Data Analysis Using Python and Pandas

Introduction

The objective of this project is to analyze a sales dataset using Python and the Pandas library. The project follows a step-by-step data analysis workflow, including data loading, exploration, cleaning, analysis, and reporting. The goal is to extract meaningful business insights that can support better decision-making.

Dataset Information

- Total Records: **100**
- Total Columns: **7**
- Dataset Format: **CSV**

Columns

- Date
- Product
- Quantity
- Price
- Customer_ID
- Region
- Total_Sales

Project Workflow

Day 1: Setup & Load Data

- Imported the Pandas library.
- Loaded the sales dataset from a CSV file.
- Verified that the dataset was successfully imported.

Day 2: Explore Data

- Checked the dataset shape.
- Displayed column names.
- Examined data types using `df.info()`.
- Reviewed the dataset structure.

Day 3: Clean Data

- Checked for missing values.
- Filled missing values where necessary.
- Removed duplicate records.

- Prepared the dataset for analysis.

Day 4: Sales Analysis

The following business metrics were calculated:

- **Total Revenue:** 12,365,048
- **Best Performing Product:** Laptop
- **Total Quantity Sold:** Calculated by summing the `Quantity` column.

Revenue was calculated using:

$$\text{Revenue} = \text{Quantity} \times \text{Price}$$

Day 5: Report Generation

The analysis results were summarized into a report containing:

- Dataset overview
- Cleaning summary
- Key business metrics
- Final insights

Key Insights

- The dataset contains **100 sales records** with **7 columns**.
- No duplicate records were found after data cleaning.
- **Laptop** generated the highest revenue among all products.
- The total revenue generated from the dataset is **12,365,048**.
- The dataset is clean and suitable for further visualization and advanced analysis.

Conclusion

This project demonstrates the complete data analysis process using Python and Pandas. It helped develop practical skills in loading datasets, exploring data, cleaning records, calculating business metrics, and generating reports. The project serves as a strong foundation for more advanced data analysis and visualization tasks.